

Project Design Phase

Solution Architecture

Date	4 JULY 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Solution Architecture:

Solution Architecture describes the overall structure of the Credit Card Approval Prediction system and illustrates how different components interact to generate accurate predictions. The architecture begins with the user entering transaction details through the web interface, followed by input validation and preprocessing. The processed data is passed to the trained Random Forest machine learning model, which predicts whether the transaction is genuine or fraudulent. Finally, the prediction result is displayed to the user through the Flask web application. This architecture is modular, maintainable, and allows future enhancements such as cloud deployment, database integration, and real-time transaction analysis.

SOLUTION ARCHITECTURE DIAGRAM

Project: Credit Card Approval Prediction using Machine Learning
Deployment Environment: Local Flask Server

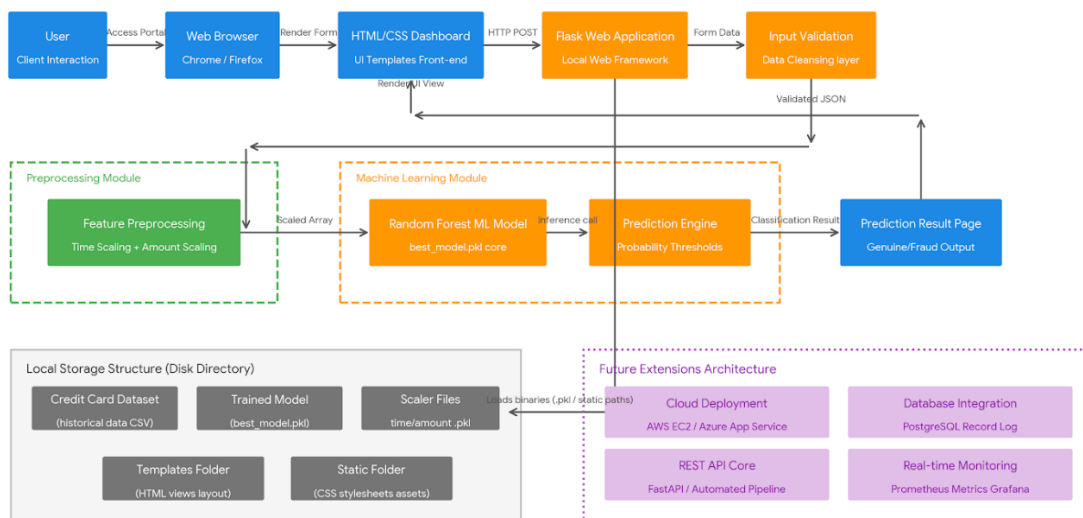


Figure 1: Architecture and data flow of Credit Card Approval Prediction

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>